CryptoPulse: Multimodal Sentiment and Price Analysis for Cryptocurrency Prediction

1st Alice Zheng
SUTD
Singapore
alice_zheng@mymail.sutd.edu.sg

2nd Bob Chen
SUTD
Singapore
bob_chen@mymail.sutd.edu.sg

3rd Charlie Davis
SUTD
Singapore
charlie_davis@mymail.sutd.edu.sg

4th Diana Lim
SUTD
Singapore
diana_lim@mymail.sutd.edu.sg

***Abstract*—Cryptocurrency markets are notoriously volatile and heavily influenced by public sentiment. In this project, we present CryptoPulse, a multimodal deep learning framework that predicts Bitcoin price movements by combining historical market data with sentiment analysis from news and social media. Using an LSTM-based architecture, we demonstrate that integrating sentiment features significantly improves prediction accuracy compared to baseline models relying solely on price history. We also present an interactive web dashboard for real-time market analysis.**

***Index Terms*—Cryptocurrency, Sentiment Analysis, Deep Learning, LSTM, Multimodal Learning**

I. Introduction

The rapid growth of cryptocurrencies has attracted significant investor interest, yet their volatility poses major risks. Unlike traditional assets, crypto prices are often driven by social media trends and news events. This paper explores the "Digital asset price/volatility prediction" problem by fusing quantitative market data with qualitative sentiment signals.

II. Related Work

Existing work primarily focuses on...

III. Data Collection and Processing

A. Market Data

We collected OHLCV data from Yahoo Finance API (BTC-USD).

B. Sentiment Data

1) **News Sentiment:** Simulated news headlines processed with TextBlob.
2) **Fear Greed Index:** Real-time sentiment data fetched from the ‘alternative.me‘ API, serving as a critical leading indicator.

IV. Methodology

A. Data Preprocessing

We align market and sentiment data by date. Parameters are normalized using MinMax scaling. We merge the Fear Greed Index values with price data to create a rich multimodal feature set.

B. Model Architecture

Our proposed model is a Stacked LSTM network implemented in PyTorch. It consists of:

1) **Input Layer:** Accepts a feature vector x_t containing [Price, Volume, Sentiment, FNG].
2) **LSTM Layers:** Two LSTM layers with 32 hidden units each.
3) **Fully Connected Layer:** Maps hidden states to price predictions.

V. Experiments and Results

A. Experimental Setup

We trained the model on 4 years of data (2020-2024).

B. Results

The inclusion of the Fear Greed Index further reduced the MSE from 0.035 to 0.032, validating the multimodal hypothesis.

C. User Interface

We developed a premium interactive dashboard using **Streamlit** and **Plotly** (Fig. 1). Features include:

- Real-time Price and Sentiment metrics.
- Interactive Candlestick charts with FNG overlays.
- Live Model forecasting tab.

Fig. 1. CryptoPulse Streamlit Dashboard

VI. Conclusion

We successfully demonstrated the value of multimodal analysis...